

# The Arrow's Deep Field: Dark-Matter Phenomenology from the Khronon, and the Unification of Event-Density Gravity

Allen Proxmire

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## Abstract

GR-II identified Event Density's gravitational class as **khronometric**: Einstein's tensor sector plus one scalar — the khronon, the substrate's arrow of time made dynamical, anchored to the cosmological rest frame. This paper shows that the same scalar naturally carries the **dark-matter phenomenology** of ED's published gravity arc, unifying the corpus's two gravity programs into one theory. The construction has one missing span and two built ends. The ends: the khronon (GR-II), and the MOND sector —  $a_0 = cH_0/(2\pi)$  from the cosmic decoupling surface (Paper\_029), the Combination Rule  $a = \sqrt{(a_N a_0)}$  (Paper\_030), BTFR (Paper\_031), the MOND field equation (Paper\_036). The span: the khronon's kinetic structure. Promoting the khronometric acceleration term to a function  $W(A^2)$  of the congruence acceleration — the established khronometric-MOND structure — the static weak field reduces to the **modified Poisson equation in the metric potential itself**,  $\nabla \cdot [\mu_{\text{tot}} \nabla \Phi] = 4\pi G\rho$  with  $\mu_{\text{tot}} = 1 + c'_{1W}$ . The high-acceleration limit is GR-I, untouched (commitment-linearity  $\implies \mu \rightarrow 1$ ); the deep-infrared limit is **uniquely fixed by matching Paper\_030** ( $\mu_{\text{tot}} \rightarrow x$ , forced-given-030), embedding the Combination Rule and BTFR relativistically — at the surfaced cost of an **infrared cancellation** (the khronon takes over the static constraint below  $a_0$ ), standard in this class. Statics isolate the construction:  $\theta = \sigma = 0$  for the aligned khronon, so the MOND carrier is the *unique* active foliation term, and the ED identity  $a_i = \frac{1}{2}\partial_i \ln b$  reads the whole sector in substrate language — *the deep field switches on where bandwidth gradients fall below the cosmic rate*. **The historical kill-checks pass structurally**: lensing tracks the MOND potential with **no added vector field** (the modification is metric-borne; the conformal-coupling disease TeVeS patched never arises; slip is  $O(\Phi)$ -suppressed); no new ghost (the  $W$ -sector adds no time derivatives in unitary gauge); the classic AQUAL ellipticity conditions hold; screening is PPN-safe in the linear khronometric corner. The one soft spot — the  $A \rightarrow 0$  degeneracy — is resolved without retrofit: the degenerate point is **exact Minkowski vacuum, which is not in ED's state space** (the substrate always ticks), the pure- $A^2$  form is a truncation whose generic completion  $\mathcal{W}(A^2, \Theta)$  is healthy at the physical cosmological point, and an **orthogonality theorem** ( $\theta \equiv 0$  in statics) sequesters the regulator family from everything tested. The khronon's cosmological face ties the arc to SCBU — **one scale,  $H_0$ , in four roles** (background rate,  $a_0$ , horizon locus,  $R_H$  boundary) — and its infrared vacuum term is naturally of order  $\rho_\Lambda \sim H_0^2 c^2/G$ , the observed dark-energy order, read as single-scale consistency (form-tier only). The result: **one khronometric gravity** — Einstein at high accelerations, MOND in the deep field, dark energy's order carried — in which “dark matter” is *the deep-infrared regime of the arrow's own*

*field*. No particle, no added vector, no new primitive. Clusters and the CMB remain owed.

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## 1. Introduction

### 1.1 What this paper does

ED’s gravity corpus contained two programs. The published arc (Papers 025–038) derived Newton, the transition acceleration  $a_0 = cH_0/(2\pi)$ , the Combination Rule, and BTFR — the dark-matter-free account of galactic dynamics — covariantized in Paper\_033 on a *postulated* metric with a *postulated* scalar. The post-pivot arc (GR-I/II) *derived* the metric (the bandwidth metric, with the weak-field Einstein tests) and *derived* the scalar (the khronon: the arrow made dynamical). This paper supplies the one missing identification: **the postulated scalar of the MOND arc is the derived scalar of the curvature arc**. It proceeds in four steps: the mapping (§3), the static weak-field reduction with the GR and MOND limits (§4), the lensing and viability checks (§5–§6), and the cosmological regulator with the SCBU tie-in (§7).

Three orienting statements, up front:

- **The hinge.** The paper’s only derivational hinge is the deep-infrared matching to Paper\_030’s geometric-mean law, which **uniquely fixes** the non-analytic term in  $W'$ . No other freedom is exercised anywhere in the construction; everything else is inherited, derived, or labeled a family.
- **The economy.** No new primitive, field, or symmetry is introduced. The khronon was *counted* by GR-II’s mode analysis before MOND was on the table; this paper spends a field the theory already had.
- **What survives (preview).** The three historical kill-checks of relativistic MOND all pass structurally: **lensing** tracks the MOND potential with no added vector (§5); **no new ghost** arises (§6.1); **solar-system screening** is PPN-safe (§6.3). The one soft spot — the deep-vacuum degeneracy — is resolved at the level of ED’s state space and EFT generality (§7). And the cosmological regulator is a *family, sequestered from every static test*: its role is to make the theory well-posed on FRW backgrounds, not to influence galactic phenomenology (§7.3).

### 1.2 Why this matters

Three reasons. First, **unification**: the corpus’s two gravity lines fuse into one theory with one extra field — and that field was not added for the purpose; it was *counted* by GR-II’s mode analysis before MOND was on the table. Second, **the dark-matter claim sharpens**: “no dark matter; modified dynamics below  $a_0$ ” becomes “the galactic phenomenology is the deep-infrared regime of the preferred-foliation scalar” — a specific field with an independent origin (the arrow), independent constraints (GW170817, PPN, pulsars), and independent predictions (a breathing GW mode; wide-binary deviations). Third, **the historical failure modes are addressed structurally, not by patches**: the constructions that killed earlier relativistic MOND (conformal couplings blind to lensing; added vectors; fine-tuned  $a_0$ ) are each avoided by something ED already had.

**Why the unification is nontrivial.** Four scales that have no business coinciding — the rate of the arrow’s cosmological flow, the acceleration at which galactic dynamics turn non-Newtonian,

the energy density of the vacuum, and the locus where horizons form — are here carried by **one scalar field with one background scale**. In any framework where these are independent sectors, their numerical coincidences ( $a_0 \approx cH_0/2\pi$ ;  $\rho_\Lambda \sim H_0^2 c^2/G$ ) are unexplained accidents. In this construction they are the *same number wearing four hats*, because the khronon has only one number to offer.

### 1.3 How this fits the arc

- **Papers 027–031, 036:** Newton;  $a_0$ ; Combination Rule; BTFR; the MOND field equation. (*the deep-field end — inherited*)
- **Paper\_033:** scalar-tensor covariantization on a postulated metric and scalar. (*fused here: the scalar is identified*)
- **Paper GR-I:** the bandwidth metric; weak-field Einstein tests. (*the high-acceleration end — inherited*)
- **Paper GR-II:** the khronometric class; the khronon;  $c_T = c$ ; Lorentz safety. (*the field itself — inherited*)
- **Paper KM-I (this paper):** the unification — the khronon’s deep field is the MOND sector.

This completes the second of the two reconciliation items GR-I §7 declared open (the khronon $\leftrightarrow$ MOND relation); the first (the bandwidth $\leftrightarrow$ strain/flow substrate-route identification) remains open.

### 1.4 Conventions and regime of validity

Signature  $(-, +, +, +)$ ;  $c = 1$  where convenient. The derivations are **static, weak-field, quasi-spherical**, in the **continuum (DCGT) limit**; the cosmological statements (§7) are FRW-background structural results. **No strong-field, no time-dependent-lensing, no cluster, and no CMB claims are made.** Exact numerical coefficients in the reductions are convention-dependent and carried schematically ( $c_1$ -type constants of order unity); the *structures* and *limits* are the results.

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## 2. Primitive Inputs

**Substrate primitives (postulated, Paper\_087):** P11/P13 (the arrow and the tick — the khronon’s origin, via GR-II), P04 (bandwidth — the metric and lapse, via GR-I), P02/P05 (adjacency/transport). The  $\Sigma$ -selection’s orientation-blindness and P-Commitment-Linear (GR-I) enter through the inherited results.

### Inherited results (load-bearing):

- **GR-I:**  $g_{ij} \sim b^{-1}$ ,  $N^2 \sim b$ , the weak-field Schwarzschild metric; P-Commitment-Linear  $\implies$  the Einstein branch at high accelerations.
- **GR-II:** the khronometric class; the khronon  $T$  with  $u_\mu = \partial_\mu T / \sqrt{X}$  hypersurface-orthogonal; the preferred frame = the cosmological rest frame;  $c_T = c$ ; universal Lorentz violation.
- **Paper\_029:**  $a_0 = cH_0/(2\pi)$  (value anchored to  $H_0$ ).
- **Paper\_030:** the Combination Rule  $a = \sqrt{(a_N a_0)}$  (the substrate’s own deep-field law, via the P14 bilocal coupling).

- **Papers\_031/036:** BTFR; the MOND field equation (the flat-space form this paper’s static limit must and does land on).

**External dictionary (inherited mathematics/physics, labeled I):**

- **Khronometric/aether-MOND** (Blanchet–Marsat 2011; Zlosnik–Ferreira–Starkman 2007): a preferred-foliation scalar with non-canonical *acceleration* kinetics yields the MOND modified-Poisson equation in the static weak field, with the interpolation given by the kinetic function’s derivative. The class’s reported stability corners are cited as reported.
- **Classical AQUAL well-posedness** (Bekenstein–Milgrom): ellipticity conditions  $\mu > 0$ ,  $d(x\mu)/dx > 0$ .

**No new primitive is introduced; no primitive forcing is invoked.** The paper’s single structural commitment beyond inheritance is the *typing* of the kinetic sector (the acceleration invariant, §3.2) — which is argued, not assumed, since the alternative typing fails independently.

## 2.5 Load-Bearing Step Audit

*Status legend:* **I** = inherited (from a cited paper or established external result); **D** = derived here; **D-structural** = derived from structure alone, no model details; **conditional** = derived given a labeled input; **family** = an honestly-unfixed functional freedom; **form-tier** = an order-of-magnitude/structural consistency, no value claimed; **open** = declared open.

| Step                                                                                             | Status              | Source / justification                                                   |
|--------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------|
| Khronon exists; class is khronometric; cosmic-frame anchored                                     | I                   | GR-II                                                                    |
| Metric/lapse from bandwidth; Einstein branch at high $A$                                         | I                   | GR-I                                                                     |
| $a_0 = cH_0/2\pi$ ; Combination Rule; BTFR; MOND field eq                                        | I                   | Papers 029/030/031/036                                                   |
| Khronometric-MOND dictionary ( $W(A^2) \Rightarrow$ modified Poisson)                            | I                   | Blanchet–Marsat; ZFS                                                     |
| Kinetic invariant = the congruence <b>acceleration</b> (not $(\partial T)^2$ )                   | <b>D</b>            | §3.2 — the $(\partial T)^2$ typing is structurally inert or lensing-dead |
| Statics isolate the $W$ -sector ( $\theta = \sigma = 0$ ; khronon aligns)                        | <b>D</b>            | §4.1 — kinematics of static congruences                                  |
| ED identity $a_i = \frac{1}{2}\partial_i \ln b$                                                  | <b>D (identity)</b> | §4.1 — via GR-I’s $N^2 \sim b$                                           |
| Modified Poisson                                                                                 | <b>D</b>            | §4.2 — form-level; constants conventional                                |
| $\nabla \cdot [\mu_{\text{tot}} \nabla \Phi] = 4\pi G\rho$ ,<br>$\mu_{\text{tot}} = 1 + c'_{1W}$ |                     |                                                                          |
| GR limit ( $W' \rightarrow \alpha$ small $\Rightarrow \mu \rightarrow 1$ ; GR-I untouched)       | <b>D</b>            | §4.3                                                                     |

| Step                                                                                  | Status                           | Source / justification                                            |
|---------------------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------|
| Deep-IR limit $\mu_{\text{tot}} \rightarrow x \implies$<br>Combination Rule + BTFR    | <b>D conditional on I-030</b>    | §4.4 — <b>forced-given-030</b> ;<br>includes the IR cancellation  |
| The IR cancellation<br>( $W' \rightarrow -1/c_1 + x/c_1$ )                            | <b>surfaced cost</b>             | §4.4 — standard-in-class;<br>stress-tested §6                     |
| Lensing tracks the MOND<br>potential; no vector needed                                | <b>D</b>                         | §5 — metric-borne<br>modification; slip<br>$O(\Phi)$ -suppressed  |
| No new ghost ( $W$ -sector adds<br>no time derivatives)                               | <b>D-structural</b>              | §6.1 — unitary-gauge<br>counting                                  |
| AQUAL ellipticity across the<br>interpolation                                         | <b>D</b>                         | §6.2 — classic conditions;<br>deep IR passes                      |
| Screening PPN-safe;<br>interpolation family<br>Cassini-constrained                    | <b>D / I</b>                     | §6.3 — linear-khronometric<br>corner; ephemeris bounds            |
| $\tau$ -mode = the breathing<br>scalar; benign as screened                            | <b>D</b>                         | §6.4                                                              |
| ( $A, \theta$ ) = (0, 0) outside ED's<br>state space (no Minkowski<br>vacuum)         | <b>D-structural</b>              | §7.1 — P11/P13; the<br>khronon's background is the<br>Hubble flow |
| Degeneracy = truncation<br>artifact; $\mathcal{W}(A^2, \Theta)$ generic<br>completion | <b>D (EFT-generality)</b>        | §7.2 — regulator <b>family</b> ,<br>labeled                       |
| Orthogonality: tested sector<br>invariant under the regulator                         | <b>D</b>                         | §7.3 — $\theta \equiv 0$ in statics                               |
| One scale in four roles; SCBU<br>carrier                                              | <b>structural identification</b> | §7.4                                                              |
| $\rho_\Lambda \sim H_0^2 c^2 / G$ vacuum-term<br>resonance                            | <b>form-tier only</b>            | §7.4 — no value derived;<br>V1-reconciliation open                |
| Čerenkov / BBN / FRW<br>stability                                                     | <b>filters on the family</b>     | §7.5                                                              |
| Primitives-level origin of the<br>deep-IR branch                                      | <b>OPEN — guarded</b>            | preamble 2                                                        |
| Clusters / CMB                                                                        | <b>NOT ADDRESSED</b>             | preamble 8                                                        |

All steps I, D, structural, conditional, family, form-tier, open, or not-addressed. *The single derivational hinge is forced-given-030; the two honest families (interpolation; regulator) are sequestered and independently constrained.*

### 3. The Mapping: One Missing Span, Two Built Ends

#### 3.1 The ends

**The field (GR-II).** ED's arrow lives in the law, forcing a preferred foliation; the propagating content is 2 tensor modes + 1 scalar — the khronon, whose background is cosmic time and

whose background rate is the Hubble rate.

**The law and the scale (Papers 028–031, 036).**  $a_0 = cH_0/(2\pi)$  from the decoupling surface; the Combination Rule  $a = \sqrt{(a_N a_0)}$ ; BTFR; and the flat-space MOND field equation this construction’s static limit must reproduce.

### 3.2 The span, and its typing

The missing piece is the khronon’s kinetic structure. Its **typing is argued, not chosen**: a kinetic function of  $(\partial T)^2$  fails twice over — in statics  $(\partial T)^2 \approx 1 - 2\Phi - |\nabla\tau|^2$  depends on the *potential*, not its gradient (yielding screening-type physics, never  $\mu(|\nabla\Phi|/a_0)$ ), and its conformally-coupled variant is the historical RAQUAL construction that fails lensing. The correct carrier is the **acceleration of the khronon congruence**,  $a_\mu = \perp_\mu^\nu \nabla_\nu \ln \sqrt{X}$  — and this stays inside the foliation EFT GR-II already counted: the general quadratic sector is  $\lambda\theta^2 + \beta\sigma_{\mu\nu}\sigma^{\mu\nu} + \alpha a_\mu a^\mu$ , and the MOND extension promotes **only the acceleration term**:

$$S_T = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[ R + \lambda\theta^2 + \beta\sigma_{\mu\nu}\sigma^{\mu\nu} + a_0^2 W(A^2) \right], \quad A^2 = \frac{a_\mu a^\mu}{a_0^2},$$

with matter coupled to  $g_{\mu\nu}$  **only** (universal coupling — in ED, matter *is* substrate content; there is no conformal matter frame to mis-couple).  $W$  is carried as a general function; its two asymptotics are fixed in §4, the region between them is an honest family.

**The scale’s structural carrier.** The khronon’s vacuum is the Hubble flow, so the only rate available to normalize its local gradients is the cosmic rate: the comparison “local acceleration against  $cH_0$ ” is not a model choice but the field’s only option.  $a_0$  **is the khronon’s background scale** — the Milgrom coincidence  $a_0 \approx cH_0/2\pi$  acquires a carrier (the coefficient  $1/2\pi$  stays inherited from Paper\_029).

## 4. The Static Weak-Field Reduction

With the typing fixed (§3.2), the reduction proceeds in three steps: **alignment** (the khronon locks to static time), **isolation** (statics switch off every foliation term except the MOND carrier), and **reduction** (the modified Poisson equation, with its two limits).

### 4.1 Alignment, isolation, and the Bandwidth–Acceleration Lemma

Static, time-reversal-symmetric sources align the khronon ( $T = t$ ; the misalignment  $\tau$  is odd under the symmetry), making its congruence the static observers’, whose kinematics are textbook:  $\theta = 0$ ,  $\sigma = 0$ ,  $a_i = \partial_i \ln N$ . Two consequences:

1. **Statics isolate the MOND carrier.** The  $\lambda$  and  $\beta$  sectors are inert ( $\theta = \sigma = 0$ ); the acceleration function is the *unique* active foliation term in statics. The MOND modification and the static modification are the same term — by kinematics, not tuning. (The  $\beta$ -sector governs the tensor speed:  $c_T = c$  from GR-II is untouched by  $W$ .)
2. **The Bandwidth–Acceleration Lemma.** GR-I derived  $N^2 \sim b$ , so

$$a_i = \partial_i \ln N = \frac{1}{2} \partial_i \ln b :$$

**the khronon's acceleration is the logarithmic bandwidth gradient.** This identity — referenced as such below — is the substrate reading of the entire MOND sector: the deep field switches on *where bandwidth gradients fall below the cosmic rate*.

## 4.2 The modified Poisson equation

With  $g_{00} = -(1 + 2\Phi)$  and  $A^2 = |\nabla\Phi|^2/a_0^2$  at leading order, varying the action gives (schematic order-unity constants flagged as conventional):

$$\nabla \cdot \left[ \mu_{\text{tot}} \left( \frac{|\nabla\Phi|}{a_0} \right) \nabla\Phi \right] = 4\pi G\rho, \quad \mu_{\text{tot}}(x) = 1 + c_1 W'(x^2)$$

— the AQUAL form, **in the metric potential itself** (not an auxiliary scalar). Here and throughout,  $x \equiv |\nabla\Phi|/a_0$ . The  $\Psi$ -equation receives the khronon's stress, whose anisotropic part is second order in the potentials (§5).

## 4.3 The GR limit

At high accelerations  $W' \rightarrow \alpha$ , a small constant — the standard quadratic khronometric coupling, PPN-bounded. Then  $\mu_{\text{tot}} \rightarrow 1$ ,  $\Phi = \Psi$  at leading order, and **GR-I's weak-field Schwarzschild sector — the factor-of-two bending, redshift, precession — is untouched**. The residual  $O(\alpha)$  preferred-frame effects are GR-II §8's standing  $\alpha_1, \alpha_2$  front, unchanged. ED-side, this limit is enforced: P-Commitment-Linear demands the Einstein branch at strong gradients — the screening regime is GR-I, not an option.

## 4.4 The deep-IR limit — forced-given-030, with the cost stated

Matching the corpus's derived deep-field law ( $\mu_{\text{tot}} \rightarrow x$  so that, by Gauss-law integration,  $|\nabla\Phi| = \sqrt{a_N a_0}$ ) fixes the deep-IR kinetic behavior **uniquely**:

$$W'(x^2) \xrightarrow{x \rightarrow 0} -\frac{1}{c_1} + \frac{x}{c_1} + \dots$$

Two pieces with two statuses. The non-analytic linear piece is the MOND carrier — the  $X^{3/2}$ -type term, forced-given-030. The constant  $-1/c_1$  deserves its own box:

**The IR Cancellation Requirement.** In the convention where Einstein–Hilbert keeps its standard normalization, the khronon's kinetic slope must approach the constant  $-1/c_1$  in the deep infrared — **exactly cancelling the Einstein gradient term in the static constraint below  $a_0$  and taking it over**. Three facts about it: (i) it is **standard in this class** (the negative-branch kinetic function of generalized-aether MOND); (ii) it is the construction's **only delicate point** — everything else is inheritance, kinematics, or sequestered family; (iii) it is **stress-tested in §6**, where it is shown *not* to destabilize statics (the total operator stays healthy even where  $W' < 0$ ) and *not* to introduce a ghost.

With the requirement in place, the Combination Rule and BTFR slope-4 ( $v^4 = GMa_0$ ) are **embedded relativistically** — the khronon carries Paper\_030’s law; it does not re-derive it.

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## 5. The Lensing Verdict

**The check.** Galactic lensing must track the MOND potential — the historical killer of relativistic MOND.

**Why this construction is on the right side of history.** The modification is **metric-borne**: §4.2 MONDifies the *metric potential*; matter and light couple universally to  $g_{\mu\nu}$ ; there is no conformal scalar sector. RAQUAL failed lensing because its scalar coupled conformally and conformal factors do not deflect light — the *same* conformal-blindness fact GR-I §6 used to exclude Nordström. TeVeS’s vector field was a patch for that disease; **the khronon never has the disease the patch was for.**

**The slip.** The khronon’s anisotropic stress is  $\sim W'a_{\langle i}a_{j\rangle} = O(|\nabla\Phi|^2)$  — second order in the potentials; even deep in the MOND regime its ratio to the leading terms is  $O(\Phi) \sim 10^{-6}$  for galaxies. Hence  $\Phi = \Psi [1 + O(\Phi)]$ , both potentials are the MONDified potential, and the deflection is  $\propto 2 \int \nabla_{\perp} \Phi_{\text{MOND}}$ :

**Galactic lensing tracks the MOND potential, with no added vector field.**

The kill-check passes at leading order — for the same structural reason ED’s gravity produced the factor of two: space and time potentials move together.

Caveats: leading-order, static; cluster-scale lensing inherits the MOND shortfall (preamble 8); the slip suppression presumes the healthy perturbations §6 addresses.

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## 6. Viability: Ghosts, Ellipticity, Screening, and the $\tau$ -Mode

### 6.1 No new ghost (structural)

In unitary gauge  $a_i = \partial_i \ln N$  — spatial gradients of the lapse only — so the  $W$ -sector contributes **no time derivatives at any order**. The scalar’s time-kinetic structure is the  $\lambda, \beta$  sector, untouched by the MOND generalization: **the no-ghost condition of the unified theory is the linear khronometric one**. (The  $W$ -sector can feed the effective kinetic matrix through constraint elimination; that feedback is exactly the §7 degeneracy item, not a second pathology.)

### 6.2 Static well-posedness (the classic conditions)

The perturbed static operator has the AQUAL coefficients; the Bekenstein–Milgrom conditions  $\mu_{\text{tot}} > 0$  and  $d(x\mu_{\text{tot}})/dx > 0$  hold across the interpolation — deep IR ( $\mu = x$ ) passes both. **The infrared cancellation does not destabilize statics:  $W' < 0$  there, but the *total* operator is healthy.**

### 6.3 Screening and the constrained family

Screening is the smallness of the residual, not Vainshtein: at solar accelerations ( $x \sim 10^7 \sim 10^8$ ) the force correction is the saturation tail plus  $O(\alpha)$  preferred-frame terms. Two channels, kept



distinct: the  $\alpha_1, \alpha_2$  **channel** (the linear corner; pulsar-tested; the standing GR-II falsification front, unchanged) and the **interpolation-tail channel**, where Cassini-class ephemeris bounds **exclude slow-saturating families** (inverse-power tails at the edge of exclusion; exponential tails survive). The interpolation is thereby a *constrained* family — measured, not protected.

## 6.4 The $\tau$ -mode

In unitary gauge the khronon perturbation is eaten: it **is** the breathing scalar GR-II already counted — not a new player. Benign as screened at high accelerations (dipole radiation suppressed with the scalar charge); its deep-IR dynamics (bars, mergers, satellites, wide binaries) are the theory's *predictive* sector. Its only pathology channel is the §7 degeneracy — one soft spot, one address.

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## 7. The Regulator and the Cosmological Face

### 7.1 The degenerate point is not in ED's state space

The pure- $A^2$  sector degenerates as  $A \rightarrow 0$  (vanishing stiffness, slowing modes) — but the configuration  $(A, \theta) = (0, 0)$  at which the structure fails is **exact Minkowski vacuum, eternal and inert** — and ED's substrate forbids exact Minkowski vacuum: **the tick never halts**. The khronon's background is the Hubble flow; commitment does not cease (P13, P11). The degenerate point sits on the excluded boundary of the substrate's state space. This is a domain observation, not a modification of  $W$  — and it is one of the construction's strongest structural facts: *the substrate's refusal to ever stop ticking is what protects its gravity theory*.

### 7.2 The truncation diagnosis and the regulator family

$W(A^2)$  was a truncation: the foliation EFT's invariants are  $a, \theta, \sigma$ , and nothing singles out the acceleration as the sole argument of the IR structure. The general sector  $\mathcal{W}(A^2, \Theta)$  ( $\Theta = \theta/3H_0$ ) is **non-degenerate at the physical near-vacuum** ( $A \rightarrow 0, \Theta \sim 1$ ), where the expansion supplies the stiffness. The  $\Theta$ -form is **not forced** — the **regulator family**, labeled.

### 7.3 The orthogonality theorem

$\theta \equiv 0$  for static congruences, so **every  $\Theta$ -term vanishes identically in statics: the entire tested sector of this paper (§4–§6) is invariant under the regulator choice**. The family is sequestered — it cannot be tuned to help the galactic physics (no leak-back), and the galactic physics cannot vouch for it (no protection). Two independently falsifiable sectors.

### 7.4 The SCBU tie-in, and one tiered resonance

The khronon has exactly one background scale, and it appears in four roles: the background rate ( $H_0$ ), the transition acceleration ( $a_0 = cH_0/2\pi$ ), the horizon locus ( $b \rightarrow 0$ , GR-I), and SCBU's unified boundary ( $R_H = c/H_0$ ). **One scale, four roles: the SCBU boundary acquires a dynamical carrier — its field is the khronon**. And the IR sector's vacuum term is naturally of order

$$\rho_\Lambda \sim \frac{a_0^2 c^2}{G} \sim \frac{H_0^2 c^2}{G}$$

— the observed dark-energy order: the long-noticed  $\Lambda \sim a_0^2$  coincidence read as **single-scale consistency** (the khronon’s only scale is the cosmic rate, so its vacuum term can only be that size). Tiered hard: form/order-of-magnitude only; the  $\mathcal{O}(1)$  constant is not derived;  $\Lambda \sim H_0^2$  is a consistency relation, not a prediction; and the reconciliation with the corpus’s V1-backreaction  $\Lambda$  (Paper\_038.5) is open.

### 7.5 Filters on the family

The regulator family carries its own constraints: **BBN** (the  $\lambda\theta^2$  sector renormalizes cosmological  $G$ ; bounded), **gravitational Čerenkov** (admissible members must give  $c_s \sim \mathcal{O}(1)$  at the cosmological point — the bare truncation fails this in voids, independently confirming §7.2’s diagnosis), and **FRW stability**. Pinning a member is model-building, outside this paper — the same epistemic boundary GR-II drew for the khronon parameters.

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## 8. Position Relative to the Published Arc — the Fusion

- **Paper\_033’s scalar is the khronon.** The pre-pivot arc *constructed* a scalar  $\Phi$  on a postulated metric (both labeled postulates in its own audit); the post-pivot arc *derived* a scalar with exactly the required role and origin. The two gravity programs fuse: Paper\_036’s MOND field equation is this paper’s §4.2 static limit; Paper\_033’s covariantized equation is its covariant form, with the postulated background replaced by GR-I’s metric.
- **Paper\_033/034’s “deep-MOND superluminality cost” is the khronon’s known class character** — what makes universal horizons exist — and not a causal pathology given the foliation: a flagged cost becomes an expected feature.
- **GR-I §7’s open item #2 (khronon $\leftrightarrow$ MOND) is hereby closed;** open item #1 (bandwidth $\leftrightarrow$ strain/flow) remains.
- **“Dark matter” in ED:** the galactic phenomenology is carried by the same field that carries the arrow of time — no particle, no added vector, no new primitive; the field was counted (GR-II) before it was needed (here).

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## 9. The Wedge — Empirical Content

1. **Wide binaries near  $a_0$**  — the live observational frontier where this theory and Newtonian expectations genuinely diverge; an active measurement program, and the cleanest near-term discriminator.
2. **Dynamical MOND** — the  $\tau$ -mode’s excitation in time-dependent galactic systems is where MOND-class dynamics differ from particle dark matter; predictive territory, not yet computed.
3. **The breathing GW polarization** (GR-II’s scalar mode) — now identified as the *same* field that does the galactic work: one detection channel, two phenomenologies.

4. **The constrained families as shrinking falsifiable space** — Cassini on the interpolation; BBN/Čerenkov/FRW on the regulator: the theory’s unfixed sectors are being measured, not protected.
  5. **What is *not* claimed as a wedge:** clusters and the CMB (owed); the value of  $\Lambda$  (form-tier only); solar-system deviations beyond the standing  $\alpha_1, \alpha_2$  front.
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## 10. Falsification Criteria

### 10.1 Lensing slip in the MOND regime

**Falsifier sentence:** *Observation of a leading-order gravitational slip ( $\Phi \neq \Psi$  beyond  $O(\Phi)$  corrections) in galactic lensing relative to dynamics would falsify §5’s metric-borne mechanism — the construction has no vector field to repair it with.*

### 10.2 The Combination-Rule embedding

**Falsifier sentence:** *Galactic data inconsistent with  $a = \sqrt{(a_N a_0)}$  in the deep field (or BTFR slope-4 with the inherited  $a_0$ ) falsifies the deep-IR matching — and with it the unification, since the deep-IR form is forced-given-030.*

### 10.3 Wide binaries

**Falsifier sentence:** *Wide-binary samples near  $a_0$  showing purely Newtonian behavior at the precision where the deep-IR branch predicts deviation would falsify the unscreened sector.*

### 10.4 The interpolation family closes

**Falsifier sentence:** *Ephemeris (Cassini-class) bounds tightening to exclude all interpolation members compatible with the two fixed asymptotics would falsify the construction’s middle — there is no mechanism here to evade the static sector’s own equation.*

### 10.5 The AQUAL conditions

**Falsifier sentence:** *A demonstration that  $\mu_{\text{tot}} > 0$  or  $d(x\mu_{\text{tot}})/dx > 0$  must fail somewhere on the physical interpolation would falsify static well-posedness (§6.2).*

### 10.6 The regulator family empties

**Falsifier sentence:** *A proof that no  $\mathcal{W}(A^2, \Theta)$  member simultaneously satisfies BBN, Čerenkov, and FRW-stability filters while remaining  $\Theta$ -trivial in statics would falsify the §7 resolution of the degeneracy — returning the  $A \rightarrow 0$  soft spot to the theory’s ledger as unresolved.*

### 10.7 Preferred-frame parameters (standing)

**Falsifier sentence:** *The GR-II preferred-frame front is resolved in companion work GR-IV —  $\alpha_2 = 0$  exactly (both cones luminal) and  $\alpha_1 = -4\lambda_{\text{local}}$  density-suppressed (safe by  $\geq 70$  orders). This unification adds no protection of its own and needs none; the front stays live as a test — a measured  $\alpha_1 \gtrsim 10^{-5}$ , or any nonzero  $\alpha_2$ , kills the theory.*

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## 11. Position Statement

**Event-Density gravity is one khronometric theory, and its dark-matter phenomenology is the deep-infrared regime of the khronon — the arrow of time made dynamical.** Given the inherited ends — the khronon and its cosmic anchoring (GR-II), the metric and lapse (GR-I), the scale  $a_0 = cH_0/2\pi$  (Paper\_029) and the deep-field law (Paper\_030) — the khronometric acceleration sector with a general kinetic function reduces, in the static weak field, to the AQUAL equation in the metric potential; the high-acceleration limit is GR-I (enforced by commitment-linearity); the deep-infrared limit embeds the Combination Rule and BTFR (forced-given-030, with the infrared cancellation surfaced); lensing tracks the MOND potential with no added vector; no new ghost arises; the classic ellipticity conditions hold; screening is PPN-safe with a Cassini-constrained interpolation family; the  $A \rightarrow 0$  degeneracy is resolved at the level of ED's state space (no Minkowski vacuum) and EFT generality (the sequestered regulator family); and the khronon's cosmological face carries SCBU's one scale in four roles, with a vacuum term naturally of the observed dark-energy order (form-tier).

**What this paper claims.** The unification: the corpus's MOND sector and its curvature sector are one khronometric theory; the postulated scalar of Paper\_033 is the derived khronon of GR-II; the deep-field law is embedded, the historical kill-checks (lensing foremost) pass structurally, and the open sectors are honest, sequestered, and independently constrained families.

**What this paper does not claim.** The MOND exponent is not derived from primitives (forced-given-030);  $a_0$  and  $G$  are inherited; no value of  $\Lambda$  is derived; no interpolation or regulator member is selected; the class's stability corners are cited, not re-verified; clusters and the CMB are not addressed; and the primitives-level origin of the deep-infrared branch — *why* the substrate's response goes geometric-mean below the cosmic rate — remains the deepest declared open question.

**Series context.** Paper KM-I of the post-pivot curvature-emergence line, companion to GR-I and GR-II — and the fusion point with the published gravity arc (Papers 025–038). The remaining work is phenomenology (pin the rule; pin the families; compute the dynamical MOND sector) and the guarded origin question.

KM-I completes the unification GR-I and GR-II opened: **one scalar, one scale, one gravity.**

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## Appendix: Cross-References, Glossary, Reader Map

### Cross-references

- **Paper\_087:** position paper (13 primitives).
- **Papers\_028/029/030/031/036:** decoupling surface;  $a_0$ ; Combination Rule; BTFR; MOND field equation. (*the deep-field end*)
- **Paper\_033/034:** scalar-tensor covariantization; deep-MOND superluminality. (*fused/reframed here*)
- **Paper GR-I:** bandwidth metric; lapse; weak-field Einstein tests. (*the high-acceleration end*)

- **Paper GR-II:** the khronometric class; the khronon; GW and Lorentz safety. (*the field*)
- **SCBU:** the  $R_H = c/H_0$  boundary — here acquiring its dynamical carrier.

## Glossary

- **Khronon.** The preferred-foliation scalar (GR-II): ED's arrow made dynamical; background = cosmic time.
- **Congruence acceleration**  $a_\mu \cdot \perp \nabla_\mu \ln \sqrt{X}$ ; the kinetic invariant of the MOND sector.
- **Bandwidth–Acceleration Lemma.** In statics,  $a_i = \partial_i \ln N = \frac{1}{2} \partial_i \ln b$  — the khronon's acceleration *is* the logarithmic bandwidth gradient (via GR-I's  $N^2 \sim b$ ). The substrate reading of the MOND sector.
- $W(A^2) / \mathcal{W}(A^2, \Theta)$ . The acceleration kinetic function and its regulated completion;  $\mu_{\text{tot}} = 1 + c'_{1W}$ .
- **Forced-given-030.** The deep-IR form is uniquely fixed by matching the corpus's independently-derived Combination Rule — an embedding, not a primitives derivation.
- **The infrared cancellation.**  $W' \rightarrow -1/c_1$  as  $A \rightarrow 0$ : the khronon takes over the static constraint below  $a_0$ ; standard-in-class, the construction's delicate piece.
- **Orthogonality theorem.**  $\theta \equiv 0$  in statics  $\implies$  the regulator family is invisible to everything tested.
- **Regulator family.** The  $\Theta$ -sector of  $\mathcal{W}$ : open, sequestered, filtered by BBN/Čerenkov/FRW stability.
- **One scale, four roles.**  $H_0$  as the khronon background rate,  $a_0$ , the  $b \rightarrow 0$  horizon, and the SCBU boundary.
- **Deep field.** The khronon's  $A \ll 1$  regime — ED's account of dark-matter phenomenology.

## Reader map and open work

**Where to look next.** - *The metric and the classical tests:* GR-I. — *The class, GW speed, Lorentz safety:* GR-II. - *The deep-field laws this paper embeds:* Papers 029/030/031/036. - *The cosmological face and the boundary:* SCBU.

**Open work** (declared): the primitives-level origin of the deep-IR branch (guarded); pinning the interpolation and regulator families (model-building; Cassini/BBN/Čerenkov/FRW filters active); the dynamical MOND ( $\tau$ -mode) sector; the V1-backreaction  $\leftrightarrow$  khronon-vacuum reconciliation for  $\Lambda$ ; the bandwidth $\leftrightarrow$ strain/flow substrate-route identification (GR-I §7 item #1); clusters and the CMB (owed; address named — the  $\Theta$ -sector's cosmology).

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## End of Paper KM-I.

*Post-pivot curvature-emergence line, fusion point with the published gravity arc. The khronon — the arrow made dynamical — carries ED's dark-matter phenomenology: GR at high accelerations (enforced), the Combination Rule and BTFR below  $a_0$  (embedded, forced-given-030), lensing correct with no added vector (metric-borne), no new ghost, PPN-safe, with the  $A \rightarrow 0$  degeneracy resolved at the state-space and EFT-generalty level and the regulator sequestered by the orthogonality theorem. One scale ( $H_0$ ) in four roles ties the theory to SCBU, and the khronon's vacuum term is naturally of the dark-energy order (form-tier). No particle, no vector, no new primitive. Clusters/CMB owed; origin question guarded.*